

The role of the laryngeal articulator in synchronic and diachronic variability of word prosody
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Among the two main types of word prosody, stress shows greater phonetic variability than tone. Tone is usually realised through pitch, as reflected in its common definitions as lexicalised pitch (e.g. Hyman 2006: 229). Typical cues for stress include duration, F0, intensity, and spectral emphasis, with duration appearing as the most reliable measure across languages (Gordon & Roettger 2017: 8; van Heuven & Turk 2020: 155, 160).

Some phonetic dimensions of both stress and tone, however, remain typologically understudied, including lower vocal tract (LVT) articulations that may accompany lexical prosody (Seifart et al. 2018: e329-30; but see e.g.). Yet, some languages around the world display a wide range of such cues as duration, pitch, and LVT articulations, sometimes also vowel quality, in relation to stress and tone, with stress languages manifesting greater variability (see examples in Ivanov 1975; González 2013; Avelino Becerra, Coler & Wetzels 2016; Kuznetsova 2025).

This variability appears to be both synchronic (within a pool of observed phonetic realisations) and diachronic (regarding reconstructed paths of word prosody evolution) and is especially typical for longer segments and heavy syllable nuclei. The interaction between synchronic and diachronic variability will be exemplified with Lithuanian word prosody (based on extensive published phonetic and phonological work). Furthermore, mutual changes among quantity, pitch and LVT cues in stress languages will be illustrated with diachronic pathways identified for languages with ternary quantity in Kuznetsova (2025).

The source of this variability is explained by a comprehensive Laryngeal Articulator Model (Esling et al. 2019; Moisik et al. 2021). All pitch and LVT articulations (both laryngeal and pharyngeal) are produced by a single aryepiglottic constrictor mechanism together with associated tongue retraction and larynx raising (Esling et al. 2019: 5). Mutual synchronic and diachronic variability of pitch and LVT cues, therefore, is natural and expected in this model.

The Laryngeal Articulator Model, however, does not directly link duration to articulatory events in the lower vocal tract (Kuznetsova 2025: 349-51). As said, this variability is greater in longer segments and syllables (especially stressed). Lengthening means a longer time over which laryngeal mechanisms accompany oral postures throughout articulation, and longer movement entails more variable motion (Turk & Shattuck-Hufnagel 2020: 90-95). Greater variability implies a larger pool of phonetic output and, therefore, more diverse potential sources for sound change (Ohala 1989). In particular, for longer postures, we might expect greater changes in the degree of larynx constriction, leading to more noticeable and variable pitch and intensity falls and stronger voice quality effects than for shorter ones. LVT articulations accompanying stressed syllables are often “ballistic” (momentary, with uncontrolled output), which also contributes to increased phonetic variability. Finally, greater variability in the acoustic cues accompanying lexical stress might be observed not only with increased duration but also with stronger overall effort on stressed syllables. For example, pragmatically motivated “prosodic augmentation” (Lovick 2023: 389) on certain expressive lexemes in Shiwiar is variably expressed through longer duration, higher pitch, and more prominent LVT cues (Kohlberger 2024).

In sum, longer, more intense, and less controlled LVT articulations accompanying stress are likely responsible for this remarkable phonetic variability of durational, pitch, and LVT cues in (especially heavy) stressed syllables, which can be lexicalised in different ways across languages. Tonal articulatory targets generally appear more controlled and hence less variable in this respect than stressed syllables; however, longer segments and heavy syllable nuclei still exhibit more variability than shorter units, even in tonal languages.

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